

INFORMATION RETRIEVAL

INFORMATION RETRIEVAL: SEARCHING IN THE 21ST CENTURY

Ayşe Göker

City University London, UK

John Davies

BT, UK



WILEY

A John Wiley and Sons, Ltd., Publication

This edition first published 2009
© 2009 John Wiley & Sons, Ltd.

Except for Chapter 3, 'Multimedia Resource Discovery' © 2009 Stefan Rieger.

Registered office

John Wiley & Sons, Ltd, The Atrium, Southern Gate, Chichester, West Sussex, PO19 8SQ, United Kingdom

For details of our global editorial offices, for customer services and for information about how to apply for permission to reuse the copyright material in this book please see our website at www.wiley.com.

The right of the authors to be identified as the authors of this work has been asserted in accordance with the Copyright, Designs and Patents Act 1988.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise, except as permitted by the UK Copyright, Designs and Patents Act 1988, without the prior permission of the publisher.

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print may not be available in electronic books.

Designations used by companies to distinguish their products are often claimed as trademarks. All brand names and product names used in this book are trade names, service marks, trademarks or registered trademarks of their respective owners. The publisher is not associated with any product or vendor mentioned in this book. This publication is designed to provide accurate and authoritative information in regard to the subject matter covered. It is sold on the understanding that the publisher is not engaged in rendering professional services. If professional advice or other expert assistance is required, the services of a competent professional should be sought.

Library of Congress Cataloging-in-Publication Data

Information retrieval : searching in the 21st century / [edited by] Ayşe Göker, John Davies.
p. cm.

Includes bibliographical references and index.

ISBN 978-0-470-02762-2

1. Information retrieval. I. Göker, Ayşe. II. Davies, J. (N. John)

ZA3075.I55 2009

025.5'24 – dc22

2009025912

A catalogue record for this book is available from the British Library.

ISBN: 978-0-470-02762-2 (H/B)

Set in 9/11 Times New Roman by Laserwords Private Ltd, Chennai, India.

Printed and bound in Great Britain by Antony Rowe Ltd, Chippenham, Wiltshire.

Dedication

I would like to thank my husband, parents and wider family for all their support to my efforts to reach my full potential. This book is one of the tangible outcomes of this process. My father, in particular, would have loved to have seen it.

Ayşe Göker

Contents

Foreword	xiii
Preface	xv
About the Editors	xvii
List of Contributors	xix
Introduction	xxi
1 Information Retrieval Models	1
<i>Djoerd Hiemstra</i>	
1.1 Introduction	1
1.1.1 Terminology	1
1.1.2 What is a model?	2
1.1.3 Outline	3
1.2 Exact Match Models	3
1.2.1 The Boolean model	3
1.2.2 Region models	4
1.2.3 Discussion	5
1.3 Vector Space Approaches	5
1.3.1 The vector space model	6
1.3.2 Positioning the query in vector space	7
1.3.3 Term weighting and other caveats	7
1.4 Probabilistic Approaches	8
1.4.1 The probabilistic indexing model	8
1.4.2 The probabilistic retrieval model	9
1.4.3 The 2-Poisson model	10
1.4.4 Bayesian network models	11
1.4.5 Language models	12
1.4.6 Google's PageRank model	14
1.5 Summary and Further Reading	15
Exercises	16
References	18
2 User-centred Evaluation of Information Retrieval Systems	21
<i>Pia Borlund</i>	
2.1 Introduction	21
2.1.1 Background	21
2.1.2 Chapter outline	23
2.2 The MEDLARS Test	23

2.2.1	<i>Description of Lancaster's test of MEDLARS</i>	24
2.2.2	<i>Evaluation characteristics</i>	25
2.3	The Okapi Project	26
2.3.1	<i>The objectives of Okapi</i>	26
2.3.2	<i>Okapi at TREC</i>	27
2.3.3	<i>The impact of Okapi</i>	28
2.4	The Interactive IR Evaluation Model	28
2.4.1	<i>The cognitive IR approach</i>	29
2.4.2	<i>The three parts of the IIR evaluation model</i>	29
2.5	Summary	31
	Exercises	34
	References	35
3	Multimedia Resource Discovery	39
	<i>Stefan Rüger</i>	
3.1	Introduction	39
3.2	Basic Multimedia Search Technologies	40
3.2.1	<i>Piggy-back text retrieval</i>	41
3.2.2	<i>Automated annotation</i>	41
3.2.3	<i>Content-based retrieval</i>	42
3.2.4	<i>Fingerprinting</i>	44
3.3	Challenges of Automated Visual Indexing	44
3.4	Added Services	45
3.4.1	<i>Video summaries</i>	45
3.4.2	<i>New paradigms in information visualisation</i>	46
3.4.3	<i>Visual search and relevance feedback</i>	49
3.5	Browsing: Lateral and Geotemporal	51
3.6	Summary	55
	Exercises	57
	References	59
4	Image Users' Needs and Searching Behaviour	63
	<i>Stina Westman</i>	
4.1	Introduction	63
4.2	Image Attributes and Users' Needs	64
4.2.1	<i>Image attributes</i>	64
4.2.2	<i>Image attributes in queries</i>	67
4.2.3	<i>Attributes beyond queries</i>	69
4.2.4	<i>Image needs</i>	70
4.3	Image Searching Behaviour	71
4.3.1	<i>Search process</i>	71
4.3.2	<i>Search strategies</i>	73
4.3.3	<i>Relevance criteria</i>	76
4.4	New Directions for Image Access	77
4.4.1	<i>Social tagging</i>	77
4.4.2	<i>Images in context</i>	77
4.4.3	<i>Visualisations</i>	78
4.4.4	<i>Workspaces</i>	78
4.5	Summary	78
	Exercise	80
	References	81

5	Web Information Retrieval	85
	<i>Nick Craswell and David Hawking</i>	
5.1	Introduction	85
5.2	Distinctive Characteristics of the Web	85
5.2.1	<i>Web data</i>	85
5.2.2	<i>Web structure</i>	86
5.2.3	<i>User behaviour</i>	87
5.2.4	<i>User interaction data</i>	87
5.3	Three Ranking Problems	87
5.3.1	<i>Retrieval</i>	88
5.3.2	<i>Selective crawling</i>	88
5.3.3	<i>Index organisation</i>	89
5.4	Other Web IR Issues	90
5.4.1	<i>Stemming</i>	90
5.4.2	<i>Treatment of near-duplicate content</i>	90
5.4.3	<i>Spelling suggestions</i>	91
5.4.4	<i>Spam rejection</i>	91
5.4.5	<i>Adult content filtering – genre classification</i>	92
5.4.6	<i>Query-targeted advertisement generation</i>	92
5.4.7	<i>Snippet generation</i>	92
5.4.8	<i>Context and web information retrieval</i>	93
5.5	Evaluation of Web Search Effectiveness	93
5.5.1	<i>TREC-9 Web Track: Realistic queries, rich link structure, traditional IR task</i>	94
5.5.2	<i>Evaluation using web-specific tasks</i>	95
5.5.3	<i>Future directions for web IR evaluations</i>	95
5.5.4	<i>Comparing results lists in context</i>	96
5.5.5	<i>Evaluation by commercial web search companies</i>	96
5.6	Summary	97
	Exercises	98
	References	99
6	Mobile Search	103
	<i>David Mountain, Hans Myrhaug and Ayşe Göker</i>	
6.1	Introduction: Mobile Search – Why Now?	103
6.1.1	<i>Technological drivers</i>	103
6.1.2	<i>Predicted demand for mobile search</i>	104
6.2	Information for Mobile Search	107
6.2.1	<i>Linking information to physical space</i>	107
6.2.2	<i>The storage of information</i>	107
6.2.3	<i>The ownership of information</i>	113
6.3	Designing for Mobile Search	115
6.3.1	<i>Characteristics of mobile usage</i>	115
6.3.2	<i>Filters as a framework for mobile search</i>	116
6.3.3	<i>Manual versus automatic filtering</i>	119
6.3.4	<i>Using filters to push information</i>	120
6.4	Case Studies	121
6.4.1	<i>Oslo airport – AmbieSense</i>	121
6.4.2	<i>The Swiss Alps – WebPark</i>	123
6.5	Summary	125
	Exercises	127
	References	129

7	Context and Information Retrieval	131
	<i>Ayşe Göker, Hans Myrhaug and Ralf Bierig</i>	
7.1	Introduction	131
7.2	What is Context?	132
7.2.1	<i>Whose context?</i>	133
7.3	Context in Information Retrieval	133
7.3.1	<i>Context in the wider sense</i>	134
7.3.2	<i>Perceptions of context in related fields</i>	135
7.3.3	<i>Example: context and images</i>	135
7.4	Context Modelling and Representation	136
7.4.1	<i>Context modelling</i>	137
7.4.2	<i>User models and their relationship to context</i>	140
7.4.3	<i>Past, present and future contexts</i>	141
7.5	Context and Content	143
7.5.1	<i>Representation of context</i>	143
7.5.2	<i>Capturing context</i>	144
7.5.3	<i>Searching with context information</i>	145
7.5.4	<i>Context templates</i>	146
7.6	Related Topics	147
7.6.1	<i>Personalisation and context</i>	147
7.6.2	<i>Mobility and context</i>	148
7.7	Evaluating Context-aware IR Systems	148
7.7.1	<i>Principles of methodology</i>	149
7.8	Summary	150
	Exercises	151
	References	153
8	Text Categorisation and Genre in Information Retrieval	159
	<i>Stuart Watt</i>	
8.1	Introduction: What is Text Categorisation?	159
8.1.1	<i>Purpose of categorisation</i>	160
8.2	How to Build a Text Categorisation System	162
8.2.1	<i>The classifier component</i>	163
8.2.2	<i>The machine learning component</i>	166
8.2.3	<i>The feature selection component</i>	167
8.3	Evaluating Text Categorisation Systems	168
8.4	Genre: Text Structure and Purpose	169
8.4.1	<i>An overview of genre</i>	169
8.4.2	<i>Text categorisation and genre</i>	171
8.4.3	<i>The importance of layout</i>	171
8.5	Related Techniques: Information Filtering	172
8.6	Applications of Text Categorisation	173
8.7	Summary and the Future of Text Categorisation	174
	Exercises	175
	References	176
9	Semantic Search	179
	<i>John Davies, Alistair Duke and Atanas Kiryakov</i>	
9.1	Introduction	179
9.1.1	<i>Limitations of current search technology</i>	180
9.1.2	<i>Ontologies</i>	181
9.1.3	<i>Knowledge bases and semantic repositories</i>	182

9.2	Semantic Web	184
9.2.1	<i>Semantic web and semantic search</i>	185
9.2.2	<i>Basic semantic web standards: RDF(S) and OWL</i>	185
9.3	Metadata and Annotations	187
9.4	Semantic Annotations: the Fibres of the Semantic Web	189
9.5	Semantic Annotation of Named Entities	191
9.5.1	<i>Named entities</i>	192
9.5.2	<i>Semantic annotation model and representation</i>	192
9.6	Semantic Indexing and Retrieval	193
9.6.1	<i>Indexing with respect to lexical concepts</i>	194
9.6.2	<i>Indexing with respect to named entities</i>	196
9.6.3	<i>Retrieval as spreading activation over semantic network</i>	196
9.7	Semantic Search Tools	197
9.7.1	<i>Searching through document-level RDF annotations – QuizRDF</i>	197
9.7.2	<i>Exploiting massive background knowledge – TAP</i>	200
9.7.3	<i>Character-level annotations and massive world knowledge – KIM</i>	200
9.7.4	<i>Squirrel</i>	203
9.7.5	<i>Other approaches</i>	207
9.8	Summary	208
	Exercises	210
	References	211
10	The Role of Natural Language Processing in Information Retrieval: Searching for Meaning and Structure	215
	<i>Tony Russell-Rose and Mark Stevenson</i>	
10.1	Introduction	215
10.2	Natural Language Processing Techniques	217
10.2.1	<i>Named entity recognition</i>	217
10.2.2	<i>Information extraction</i>	218
10.2.3	<i>WordNet</i>	220
10.2.4	<i>Word sense disambiguation</i>	221
10.2.5	<i>Evaluation</i>	222
10.3	Applications of Natural Language Processing in Information Retrieval	223
10.3.1	<i>Text mining</i>	223
10.3.2	<i>Question answering</i>	224
10.4	Discussion	225
10.5	Summary	226
	Exercises	228
	References	229
11	Cross-Language Information Retrieval	233
	<i>Daqing He and Jianqiang Wang</i>	
11.1	Introduction	233
11.2	Major Approaches and Challenges in CLIR	234
11.3	Identifying Translation Units	235
11.3.1	<i>Tokenisation</i>	236
11.3.2	<i>Stemming</i>	236
11.3.3	<i>Phrase identification</i>	236
11.3.4	<i>Stop-words</i>	237
11.4	Obtaining Translation Knowledge	237
11.4.1	<i>Obtaining bilingual dictionaries and corpora</i>	237
11.4.2	<i>Extracting translation knowledge</i>	238

11.4.3	<i>Dealing with out-of-vocabulary terms</i>	238
11.5	Using Translation Knowledge	239
11.5.1	<i>Translation disambiguation</i>	239
11.5.2	<i>Weighting translation alternatives</i>	240
11.5.3	<i>Using translation probabilities in term weighting</i>	241
11.6	Interactivity in CLIR	241
11.6.1	<i>Interactive CLIR</i>	241
11.6.2	<i>Query translation in interactive CLIR</i>	242
11.6.3	<i>Document selection in interactive CLIR</i>	243
11.7	Evaluation of CLIR Systems	244
11.7.1	<i>Cranfield-based evaluation framework</i>	244
11.7.2	<i>Evaluations on interactive CLIR</i>	245
11.7.3	<i>Current CLIR evaluation frameworks</i>	245
11.8	Summary and Future Directions	245
11.8.1	<i>Current achievements in CLIR</i>	245
11.8.2	<i>Future directions for CLIR</i>	246
11.8.3	<i>Further reading</i>	246
	Exercises	247
	References	248
12	Performance Issues in Parallel Computing for Information Retrieval	255
	<i>Andrew MacFarlane</i>	
12.1	Introduction	255
12.2	Why Parallel IR?	255
12.3	Review of Previous Work	257
12.4	Distribution Methods for Inverted File Data	257
12.4.1	<i>On-the-fly distribution</i>	258
12.4.2	<i>Inverted file replication</i>	259
12.4.3	<i>Inverted file partitioning</i>	259
12.5	Tasks in Information Retrieval	260
12.5.1	<i>The indexing task</i>	261
12.5.2	<i>The probabilistic search task</i>	261
12.5.3	<i>The passage retrieval task</i>	261
12.5.4	<i>The routing/filtering task</i>	262
12.5.5	<i>The index update task</i>	262
12.6	A Synthetic Model of Performance for Parallel Information Retrieval	262
12.7	Empirical Examination of Synthetic Model	264
12.7.1	<i>Comparative results using indexing models</i>	264
12.7.2	<i>Comparative results using search models</i>	265
12.7.3	<i>Comparative results using passage retrieval models</i>	265
12.7.4	<i>Comparative results using term selection models</i>	267
12.7.5	<i>Comparative results using index update model</i>	268
12.8	Summary and Further Research	269
	Exercises	270
	References	271
	Solutions to Exercises	273
	Index	285

Foreword

In the forty years since I started working in the field, and indeed for some years before that (almost since Calvin Mooers coined the term *information storage and retrieval* in the 1950s), there have been a significant number of books on information retrieval. Even if we ignore the more specialist research monographs and the ‘readers’ of previously published papers, I can find on my shelves or in my mental library many books that attempt (probably with the IR student in mind) to construct a coherent and systematic way of defining and presenting information retrieval as a field of study and of application.

Often such a book is the work of a single author, or perhaps a pair working together. Such works can clearly have an advantage in respect of coherence; the field is necessarily presented from a single viewpoint. On the other hand, they can also suffer for the same reason. The IR field is rich (more so now than it has ever been), and it is difficult within a single viewpoint to do justice to this richness. Readers, on the other hand, have to be constructed out of the materials to hand: the published papers, each of which has taken its own view, probably with a much narrower field of vision, and different from that of the other chosen papers.

The present book attempts the tricky task of combining the breadth of vision of multiple authors with the coherence of a single integrated work. The richness of the field is apparent in the range of chapters: from formal mathematical modelling to user context, from parallel computation to semantic search.

The topics covered also vary greatly in their historical association with the field. *Categorisation*, for example, has been around as an IR technique for quite a long time – though Stuart Watt brings a new perspective. *Mobile search* (David Mountain, Hans Myrhaug and Ayşe Göker), however, is a relatively recent development. The use of formal models (*information retrieval models*, Djoerd Hiemstra) goes back almost to the beginning, as does experimental evaluation (*user-centred evaluation of information retrieval systems*, Pia Borlund), though in both cases there have been huge changes in the past decade.

This same decade has witnessed the huge growth of the World Wide Web, and the developing dominance of web search engines (*web information retrieval*, Nick Craswell and David Hawking) as the glue which holds the web together. For many people today, IR *is* web search. It is true that there has been a huge amount of influence in both directions: search engines are largely based on techniques from both the IR research community and from previous operational systems, while IR research and practice in other environments has learnt a great deal from the forcing-house that is the web search space. This dominance of the web as *the* domain of interest is well reflected in many of the chapters in the present volume.

It is important, however, to remember that IR is not all about web search, and that the web space presents both problems and opportunities which differ from those in other domains. The desktop, the enterprise, specialist collections such as scientific papers are all examples of different domains for which search functionality is a fundamental requirement. There are references to several of these throughout the book, but specific domains with their own chapters are *multimedia resource discovery* (Stefan Rüger) and *image users’ needs and searching behaviour* (Stina Westman). The user theme is taken further in the *context and information retrieval* (Ayşe Göker, Hans Myrhaug and Ralf Bierig). More generic problem areas are addressed in *cross-language information retrieval* (Daqing He and Jianqiang Wang), in *semantic search* (John Davies, Alistair Duke and Atanas Kiryakov) and in the

chapter on *natural language processing* (Tony Russell-Rose and Mark Stevenson). Finally, a chapter on *performance issues and parallelism* (Andrew MacFarlane) addresses more technical computing concerns.

Information retrieval, from being the rather arcane subject in which I did my masters degree forty years ago, has become one of the defining technologies of the twenty first century. I believe the present book does justice to this status.

Stephen Robertson
2008

Preface

This project originated during Ayşe Göker's time as Chair of the British Computer Society Information Retrieval Specialist Group (BCS IRSG) when John Davies presented the initial opportunity. The IRSG Committee took this up and as editors we broadened the contributors list.

Each project has its challenges and this book is no exception. We thank Andrew MacFarlane of City University London first and foremost for his collegiate support and superb organisational skills combined with subject knowledge. He gave us a great boost, particularly in the latter part of the project.

We would also like to thank Margaret Graham who provided initial input on the project.

We would like to thank colleagues at City University London who each contributed in significant ways: David Bawden, Jonathan Raper, Jo Wood, Jason Dykes, and Tamara Eisenschitz. Likewise, Stuart Watt, previously of Robert Gordon University and now a consultant. Last but not least, we thank Hans Myrhaug of AmbieSense for his wholehearted commitment and support. He has been willing to carry out any task required to move the project forward. We also appreciate his fantastic eye for detail, and his skills in graphic design.

Each chapter was reviewed by at least four referees looking at the subject matter, coverage, and readability. We thank them accordingly. The referees were as follows:

Leif Azzopardi, University of Glasgow; Ralf Bierig, Rutgers University; Malcolm Clark, Robert Gordon University; John Davies, BT Research; Tamara Eisenschitz, City University London; Ayşe Göker, City University London; Sarah Hinton, Wiley; Fang Huang, Robert Gordon University; Rowan January, Wiley; Paul Lewis, Southampton University; Andrew MacFarlane, City University London; Carol Peters, Italian National Research Council; Stefan Rüger, The Open University; Ian Ruthven, Strathclyde University; Tamar Sadah, Ex Libris Ltd; Aidan Slingsby, City University London; Alan Smeaton, Dublin City University; Sarah Tilley, Wiley; Fiona Walsh, Robert Gordon University.

About the Editors

Ayşe Göker

Dr. Ayşe Göker is a senior academic at City University London. Her research since the early 90s has focused on developing novel search techniques and environments, with an emphasis on personalised and context-sensitive information retrieval and management systems. These occur particularly within mobile and wireless computing, and also in bibliographic and web environments. Her skills are in identifying user needs and developing innovative systems that meet them. In international collaborations she has also been successful, with extensive experience in designing projects and managing teams to implement them. On the teaching side, Ayşe has developed course modules in information systems on several degree programmes at both postgraduate and undergraduate levels.

Ayşe is also a company co-founder of AmbieSense Ltd, a mobile information system company. This project began as the AmbieSense EU-IST project at Robert Gordon University, Aberdeen where she was a Reader and project leader. She holds a lifetime Enterprise Fellowship from the Royal Society of Edinburgh and Scottish Enterprise. More recently she was selected for and completed the Massachusetts Institute of Technology (MIT) Entrepreneurship Development Program in Boston, USA.

In her profession, she has been the Chair of the British Computing Society's Specialist Group in Information Retrieval (BCS IRSG) (2000-2005). She was recognised for the totality of her endeavours by becoming a finalist in the Blackberry Women & Technology Awards (2005) for Best Woman in Technology (Academia).

John Davies

Dr John Davies leads the Semantic Technology research group at BT. Current interests centre around the application of semantic web technology to business intelligence, information integration, knowledge management and service-oriented environments. He is Project Director of the €12m ACTIVE EU integrated project. He co-founded the European Semantic Web conference series. He is also chairman of the European Semantic Technology Conference and a Vice-President of the Semantic Technology Institute. He chairs the NESSI Semantic Technology working group. He has written and edited many papers and books in the areas of the semantic technology, web-based information management and knowledge management; and has served on the program committee of numerous conferences in these and related areas. He is a Fellow of the British Computer Society and a Chartered Engineer. Earlier research at BT led to the development of a set of knowledge management tools which are the subject of a number of patents. These tools were spun out of BT and are now marketed by Infonic Ltd, of which Dr Davies is Group Technical Advisor. Dr Davies received the BT Award for Technology Entrepreneurship for his contribution to the creation of Infonic.

List of Contributors

Ralf Bierig

Rutgers University, USA

Pia Borlund

Royal School of Library and Information
Science, Denmark

Nick Craswell

Microsoft, UK

John Davies

BT, UK

Alistair Duke

BT, UK

Ayşe Göker

City University London, UK

David Hawking

Funnelback, Australia; and
Australian National University

Daqing He

University of Pittsburgh, USA

Djoerd Hiemstra

University of Twente, The Netherlands

Atanas Kiryakov

Ontotext, Bulgaria

Andrew MacFarlane

City University London, UK

David Mountain

City University London, UK

Hans Myrhaug

AmbieSense, UK

Stephen Robertson

Microsoft Research Cambridge, UK; and
City University London, UK

Stefan Rüger

The Open University, UK;
Imperial College London, UK; and
University of Waikato, New Zealand

Tony Russell-Rose

Endeca, UK

Mark Stevenson

Sheffield University, UK

Jianqiang Wang

State University of New York at Buffalo,
USA

Stuart Watt

Information Balance, Canada

Stina Westman

Helsinki University of Technology,
Finland

Introduction

This book covers a diverse range of important themes in modern information retrieval. We first provide a general overview of information retrieval and then describe the chapters with a roadmap for guidance on the topics covered.

What is Information Retrieval?

Information retrieval, simply put, is about finding information. It encompasses the topics of *information need*, *search*, *retrieval*, and *access*. It can involve a range of content and media. Work on the storage and retrieval of data began very soon after the invention of the digital computer. The term *information retrieval* was coined around 1952, and as early as 1945, Bush (1945) was talking about a ‘device in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility’. The increasing amount and diversity of information that is available to us now, along with the challenges these bring, make this field a continually evolving and exciting area.

Information retrieval (IR) can be explained with the following typical problem (Figure 1): On the one hand, we have a person with an information need. This need, which can initially be vague, somehow needs to be articulated into a request which describes this. The request is then converted into a *search statement* or as is now more often referred to as a *query*. On the other hand, we have information stored in collections. They sit as a potentially valuable resource, but in order to be found they need to be represented somehow and then subsequently indexed. The challenge is to provide a good match between these two in order to ensure the information presented is of relevance to the person with the original query.

More specifically, information retrieval is the process of matching the *query* against the *information objects* that are *indexed*. An index is an optimised data structure that is built on top of the information objects, allowing faster access for the search process. The indexer tokenises the text (parsing), removes words with little semantic value (so-called *stop-words*), and unifies word families (so-called *stemming*). The same is done for the query as well. *Users* express their information need as a *request* (*search terms*) and it is formulated as a *query* for the retrieval system. The information system responds by *matching information objects*, which are *relevant* to this query. Information retrieval focuses on finding relevant information rather than simple pattern matching. It is also important to note that *relevance* is a subjective notion, since different users may make various judgments about the relevance or non-relevance of particular documents or information objects to given questions.

A *retrieval strategy* (model) is an algorithm and related structures that takes a query and a set of documents and assigns a *similarity measure* between the query and each document. This similarity represents relevance to the user query. Documents are then *ranked* on the basis of their similarity to the query and presented to the user. This process can be repeated and the query can be modified.

Figure 2 depicts the concepts that are usually provided by an information retrieval system.

Information retrieval has a number of models (as will be discussed in Chapter 1). Three main models are the *Boolean model*, the *Vector Space model*, and the *Probabilistic model*. The Boolean model is built on binary relevance (documents are treated either as relevant or non-relevant). It uses *exact match search strategy* based on Boolean expressions (such as AND, OR, NOT). The Vector Space model maps both query and documents in a vector space and uses spatial distance as a similarity measure.

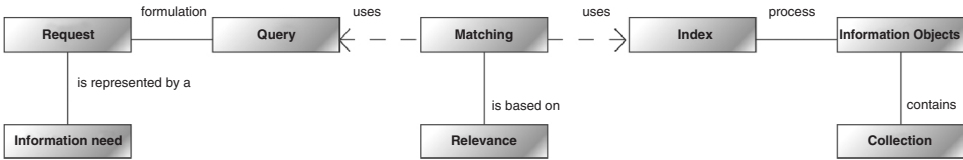


Figure 1 Information retrieval in general

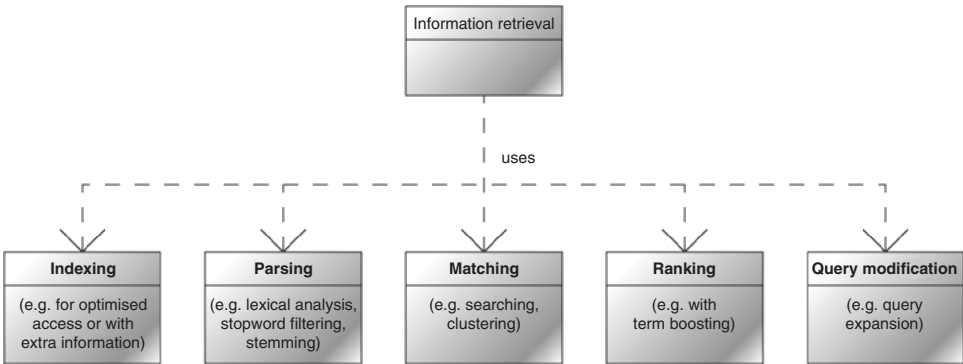


Figure 2 Information retrieval processes

On the other hand, the probabilistic model estimates document relevance as a probability using user feedback for iterative improvement. Both the vector space and the probabilistic model support ranking. There are other models that are also available, but these are a common start to the field.

Information retrieval systems tend to incorporate several activities that are also shown in Figure 2. Broadly speaking, there is *indexing*, *parsing*, *matching*, *ranking*, and *query modification*. These processes are commonly found in search engines. In practice, some of these can be brought together. For example, parsing can be part of an indexing process, and likewise the matching and ranking can be grouped under one umbrella – though not all methods of matching need to have a ranking.

Much of the research and development in information retrieval is aimed at improving retrieval efficiency (see also Chapter 2). Measures referred to as *precision* and *recall* are usually used to measure effectiveness. Precision is the ratio of the number of relevant documents retrieved in relation to the total number of documents retrieved. Recall is the ratio of the number of relevant documents retrieved in relation to the total number of relevant documents. There are other measures too, but these provide a core foundation to the field. If we are to improve information retrieval systems, then it is imperative we have some means of evaluating them against each other.

Overview of the Book

In selecting the chapters for this volume, we have attempted to offer comprehensive coverage of the main themes in modern information retrieval.

We begin with a look at the theoretical underpinnings of the field of information retrieval. As the first chapter explains, there are multiple formal IR models and each may be suited for some IR tasks and less well suited for others, and the chapter offers an excellent introduction to the principal ones in use today. With all IT systems, it is critical to know whether the system performs as intended. Believing therefore that evaluation is an important aspect of information retrieval, and further that the user view of the system and their satisfaction with the retrieved information objects are paramount, we proceed with Chapter 2, on user-centred evaluation.

An increasing proportion of information is non-textual in format: consider, for example, the emergence of *YouTube* and *Flickr* and the many similar systems as web-based sources of video and audio information. With this in mind, the next two chapters are devoted to retrieval of non-textual information resources. In a comprehensive overview of the area, Chapter 3 introduces the basic concepts of multimedia resource discovery, and looks at challenges of multimedia indexing, before discussing novel services which can be offered in the multimedia context. This chapter is followed by a more detailed look at the particular issues raised by image retrieval with a particular focus on user needs analysis (Chapter 4).

Web-based IR has of course emerged as a large area of research and development in its own right over the last 15 years, and the next chapter is devoted to this important topic. The distinctive characteristics of web IR are discussed, and the way in which IR on the web has been developed to deal with these characteristics is described. Like the web, the availability and use of mobile information environments have expanded at a dramatic rate in recent years, and mobile search is the subject of Chapter 6. The diversity and modes of use of mobile information environments globally are discussed and the distinctive features of mobile search identified. Two compelling mobile search applications from very different environments are then described. A key aspect of mobile environments is the requirement for context sensitivity, and a wider discussion on the role of context in IR is the subject of Chapter 7. The importance and role of context information in handling challenges of information search and retrieval are discussed. The capture and representation of context information is described along with examples and guidelines on how a context-aware information system or application can be developed. Consideration is then given to system evaluation in such context-aware information applications.

Unlike the topics covered by some other chapters, categorisation is an area which has been researched since the early days of IR and it continues to be an important topic. A fresh perspective is offered in Chapter 8 by Stuart Watt who interestingly focuses on the purposes of categorisation from the human problem-solving perspective and provides a framework of approaches to further work on categorisation in IR.

Chapter 9 looks at the role of semantic web technology in information retrieval. Central to this approach is the semantic analysis of the target document collection using techniques such as named entity recognition and information extraction, on the hypothesis that this deeper analysis of the target texts may offer improved IR. These semantic analysis techniques are described, followed by a survey of systems using such an approach. Following this chapter, a related topic is covered: the role of natural language technology in IR (Chapter 10). A range of natural language processing (NLP) techniques relevant to IR are described, along with how they can be applied to IR. The usefulness of their contribution to date and future prospects for NLP in IR are discussed. Cross-language IR (CLIR) is an area of increasing interest driven partly by the growing availability of non-English resources (e.g. on the web). Chapter 11 provides a comprehensive overview of the state of the art. We finish with a chapter of central importance to all types of IR: the use of parallel computing to improve the performance of what is by nature a computationally intensive task (Chapter 12). We look at different methods for distributing inverted file data on a parallel computer and the challenges of developing a model by comparing the output of a predictive model against empirical data.

The book is intended for graduate students, later stage undergraduate students, researchers and information/computer scientists, and in particular the growing number of industrial IT professionals whose role requires them to have an understanding of this increasingly important topic. Exercises have been included at the end of each chapter to help readers test and further develop their understanding of the material presented.

Reference

Bush, V. (1945). As we may think. *Atlantic Monthly* **176**(1): 101–108.